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RESEARCH ARTICLE

Enhanced Indoor Pedestrian Tracking Using UWB/PDR Fusion and NLOS Error Mitigation

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ABSTRACT Pedestrian Dead Reckoning (PDR) utilizing Inertial Navigation System (INS) technology often encounters substantial cumulative errors and random sensor-induced noise, while Ultra-Wideband (UWB) positioning can degrade severely under Non-Line-of-Sight (NLOS) conditions due to environmental factors. To overcome the above challenges, this research proposes an enhanced PDR/UWB fusion framework that significantly improves positioning accuracy. First, a novel NLOS ranging error mitigation approach is developed by integrating an Extreme Learning Machine (ELM) in conjunction with Gaussian Process Regression (GPR), with the initial parameters of the ELM optimized via the Nutcracker Optimizer. Channel Impulse Response (CIR) data collected from the UWB module serve as features, which are first processed by the ELM, and subsequently refined by the GPR, resulting in more accurate distance estimates under challenging NLOS conditions. Second, to accommodate scenarios where UWB signals are intermittently unavailable, a fusion strategy is proposed that employs a Multilayer Perceptron (MLP) neural network to predict pseudo-UWB observations from PDR data. Specifically, PDR-derived acceleration, angular velocity, and current position are fed into the MLP. When UWB measurements are absent, the MLP-predicted observations are integrated with PDR outputs using an Extended Kalman Filter (EKF), ensuring continuous and reliable localization. Experimental results demonstrate that the proposed integrated system substantially outperforms standalone UWB or EKF-based methods, achieving marked improvements in positioning accuracy and underscoring its effectiveness in delivering robust and precise indoor localization.

INDEX TERMS Nutcracker optimization algorithm, pedestrian dead reckoning, Gaussian process regression, extreme learning machine, multi-layer perceptron, extended Kalman filter, non-line-of-sight (NLOS), ultra-wideband.

I. INTRODUCTION

Indoor Positioning Systems (IPS) have become increasingly essential for providing accurate location information within buildings and urban canyons, serving as the foundation for various Location-Based Services (LBS). However, the reliability of the Global Positioning System (GPS) is significantly compromised in such environments due to signal attenuation and multipath effects [1]. While alternative technologies exist, they present their own challenges. Inertial Navigation Systems (INS) derived from Inertial Measurement Units (IMUs) suffer from cumulative sensor drift, while common Radio Frequency (RF) methods like Wi-Fi and Bluetooth Low Energy (BLE) often fail to meet the rigorous accuracy demands of modern applications [2], [3], [4].

To address these shortcomings, Ultra-Wideband (UWB) technology has emerged as a promising solution, offering high-precision distance estimation and resilience to interference [5], [6]. Despite its advantages, UWB's performance is severely degraded by Non-Line-of-Sight (NLOS) propagation, which introduces significant ranging errors. This

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necessitates a two-step process of first identifying and then mitigating NLOS errors.

Meanwhile, the use of 5G signals for positioning has attracted growing attention due to their wide deployment, high bandwidth, and potential for fine-grained localization. For instance, Zhang et al. [7] proposed a robust evolutionary particle filter (REPF) to fuse GNSS and 5G measurements, achieving strong resilience against urban NLOS conditions. To further enhance positioning geometry, Zhang et al. [8] introduced a UAV-based cooperative placement framework for optimizing multistatic 5G base stations. However, the application of 5G-based positioning in indoor environments still faces fundamental challenges, including the severe attenuation of millimeter-wave signals by walls and furniture, the limited accessibility to raw timing measurements (e.g., PRS or TOA) on commercial devices, and the high infrastructure cost associated with dense base station deployment. In contrast, UWB remains a more mature and practical solution for achieving centimeter-level indoor localization [9].

While existing machine learning models like the Extreme Learning Machine (ELM) have been applied to this problem, their performance is often capped by the limitations of conventional optimization algorithms like Particle Swarm Optimization (PSO), which can struggle with premature convergence in complex search spaces. To overcome these limitations, we propose a novel approach centered on the Nutcracker Optimization Algorithm (NOA), a recent metaheuristic that demonstrates superior balance between global exploration and local exploitation [10]. We integrate NOA with ELM (NOA-ELM) for robust NLOS identification and further enhance ranging precision by employing Gaussian Process Regression (GPR) for refined error correction.

Furthermore, relying solely on UWB is impractical, as signal loss can lead to positioning failures. To ensure continuous and reliable tracking, we fuse UWB data with Pedestrian Dead Reckoning (PDR) from an IMU. Existing fusion methods, however, have not fully explored machine learning techniques to handle intermittent UWB signal loss. We address this gap by developing a lightweight fusion framework where a Multilayer Perceptron (MLP) predicts missing UWB data, providing a simpler and more efficient alternative to complex recurrent networks like LSTM.

In summary, this paper tackles critical challenges in UWB/PDR fusion by introducing a novel, integrated solution. We combine an NOA-ELM-GPR model for advanced NLOS identification and mitigation with a recursive EKF-MLP fusion framework that ensures robust, high-accuracy localization even with sparse UWB data.

The key contributions of this paper are as follows:

(1) A novel NOA-ELM-GPR method for NLOS identification and ranging error mitigation is introduced. NOA effectively optimizes ELM parameters, improving convergence and solution quality, while GPR enhances localization precision in complex indoor scenarios.
(2) A recursive EKF-MLP fusion framework is developed which integrates PDR and UWB data, using an MLP to predict missing UWB signals and maintain continuous, reliable positioning performance even under limited UWB coverage.
(3) Through a series of experiments conducted under diverse indoor conditions, the results indicate that the proposed method achieves superior performance compared to current state-of-the-art techniques in positioning accuracy, robustness to NLOS errors, and convergence speed.

The remainder of this paper is organized as follows. Section II provides a detailed review of existing work relevant to UWB/PDR fusion and NLOS error mitigation. Section III presents the overall methodology, including the proposed NOA-ELM-GPR model and the EKF-MLP fusion framework. Section IV describes the experimental setup, evaluation protocols, and comparative results across various indoor scenarios. Finally, Section V concludes the paper and discusses future research directions.

II. RELATED WORK

This section reviews the state-of-the-art in three key areas relevant to our proposed method: NLOS identification and mitigation, intelligent optimization algorithms for positioning, and UWB/PDR data fusion techniques.

A. NLOS IDENTIFICATION AND MITIGATION

The presence of NLOS obstructions is a primary factor degrading UWB ranging accuracy [11]. Consequently, a significant body of research focuses on a two-step approach: identifying NLOS conditions and subsequently mitigating the induced errors [12].

For NLOS identification, machine learning classifiers are widely used. These methods leverage signal features to distinguish between LOS and NLOS signals, with common algorithms including Support Vector Machine (SVM) [13], XGBoost [14], Random Forest [15], and Decision Trees [16]. Another prominent technique is analyzing the Channel Impulse Response (CIR), as its multipath characteristics provide a rich source of information for classification [17], [18]. Additionally, statistical methods like distance residual analysis are employed to detect inconsistencies in range measurements that may indicate NLOS conditions [19].

Once a signal is identified as NLOS, various mitigation strategies are applied. These range from statistical approaches that use signal features to predict and correct ranging errors [20], [21], to methods that leverage deep learning on raw data for robust estimation [22], [23]. Other works formulate localization as a mathematical optimization problem to suppress NLOS-induced biases [24], [25]. A common and effective technique is residual weighting (Rwgh), where measurements suspected to be from NLOS paths are assigned lower weights within a filtering framework [26].

B. INTELLIGENT OPTIMIZATION IN INDOOR POSITIONING

To handle the complexity of tasks like parameter tuning and NLOS mitigation, bio-inspired optimization algorithms have been widely adopted in indoor positioning. ELM, known for its efficiency, is often paired with these optimizers to enhance its performance in positioning tasks ranging from geomagnetic [27] to Wi-Fi-based systems [28].

Algorithms like PSO and Grey Wolf Optimization (GWO) are frequently used to optimize ELM parameters [29], mitigate UWB NLOS errors [30], or determine optimal sensor placement to improve coverage and accuracy [31], [32]. However, a closer examination of these applications reveals domain-specific challenges. For instance, when optimizing ELM parameters, PSO's tendency to converge prematurely can limit the model's final accuracy [29]. In UWB NLOS scenarios, the complex error landscape can trap these optimizers in local minima, leading to significant, sporadic position errors [30]. These limitations highlight the need for more advanced optimization algorithms, like NOA, that can better navigate complex search spaces.

C. UWB/PDR DATA FUSION TECHNIQUES

To ensure continuous and robust positioning when UWB signals are unavailable, data fusion with self-contained sensors like IMUs is essential. PDR, derived from IMU data, provides relative position updates but suffers from unbounded error accumulation. Fusing PDR with absolute UWB measurements corrects this drift, making UWB/PDR fusion a central theme in indoor positioning research.

The most common fusion frameworks are based on Bayesian filters, particularly the KF and its variants, which typically employ a loosely-coupled architecture. In this setup, the UWB system provides position updates to correct the PDR-propagated state. A key challenge in this framework is handling erroneous UWB measurements caused by NLOS conditions. To address this, Kim and Pyun [33] proposed a lightweight KF-based system that incorporates an explicit NLOS identification module to adaptively adjust the measurement noise covariance, thereby improving robustness. Taking a similar but more dynamic approach, Guo et al. [34] developed a variable noise variance Kalman filter (VNV-KF), where an NLOS assessment function dynamically adjusts both the process and measurement noise matrices based on signal strength, reducing the filter's reliance on pre-tuned static noise parameters. Yuan et al. [35] further enhanced this concept with a robustly adaptive EKF that not only resists gross errors in UWB observations but also incorporates heading constraints from indoor maps to further refine PDR accuracy. Beyond the standard KF/EKF, other filtering techniques have also been explored. For instance, Vandermeeren and Steendam [36] compared the performance of an EKF against a Particle Filter (PF) for a shopping cart application, demonstrating that the non-parametric nature of the PF allows for the seamless integration of map information to constrain the position estimate.

While loosely-coupled systems are common, tightly-coupled architectures, which fuse raw UWB range measurements directly into the filter, offer potentially higher accuracy and robustness, especially when few UWB anchors are available. A recent advancement in this area was presented by Li et al. [37], who proposed a tightly-coupled fusion algorithm based on Factor Graph Optimization (FGO) with a sliding window. This approach moves beyond recursive filtering by optimizing a history of states and measurements, providing more stable and accurate results compared to traditional EKF-based methods. The versatility of UWB/PDR fusion is also demonstrated in minimalist hardware setups. For example, Long et al. [38] designed a system that uses just a single UWB anchor, ingeniously leveraging its ranging data to correct both the heading and step length of the PDR, proving the effectiveness of fusion even with limited infrastructure.

Despite these advancements, existing approaches still face challenges in handling intermittent UWB signal loss and can benefit from more advanced optimization and machine learning techniques, which motivates the work presented in this paper.

III. PROPOSED METHODS

This section details the proposed UWB/PDR fusion framework, which is designed to achieve robust and accurate indoor tracking by addressing key challenges in NLOS mitigation and data fusion. As illustrated in the overall system architecture in Fig. 1, the framework is built upon two primary data sources: an IMU for relative motion tracking via PDR, and an UWB system for absolute position updates. The core innovations of our approach are twofold. First, we introduce a novel NOA-ELM-GPR algorithm to accurately identify NLOS conditions and mitigate the associated ranging errors from the UWB measurements. Second, we develop a recursive EKF-MLP fusion strategy that seamlessly integrates the corrected UWB data with PDR, leveraging an MLP to predict UWB measurements during signal outages. The subsequent subsections will elaborate on the theoretical models and implementation details of each key component.

A. PREPROCESSING CIR DATA VIA OUTLIER REMOVAL FOR ACCURATE NLOS CLASSIFICATION

In UWB indoor localization systems, CIR parameters are pivotal for NLOS scenarios and correcting associated ranging deviations. Prior to NLOS classification, it is essential to perform outlier elimination to maintain data integrity. These anomalies often result from multipath propagation, signal reflection, environmental interference, or background noise, all of which can compromise localization precision and reduce model reliability. The exclusion of such outliers contributes to cleaner data inputs, thereby improving estimation accuracy and algorithmic robustness.

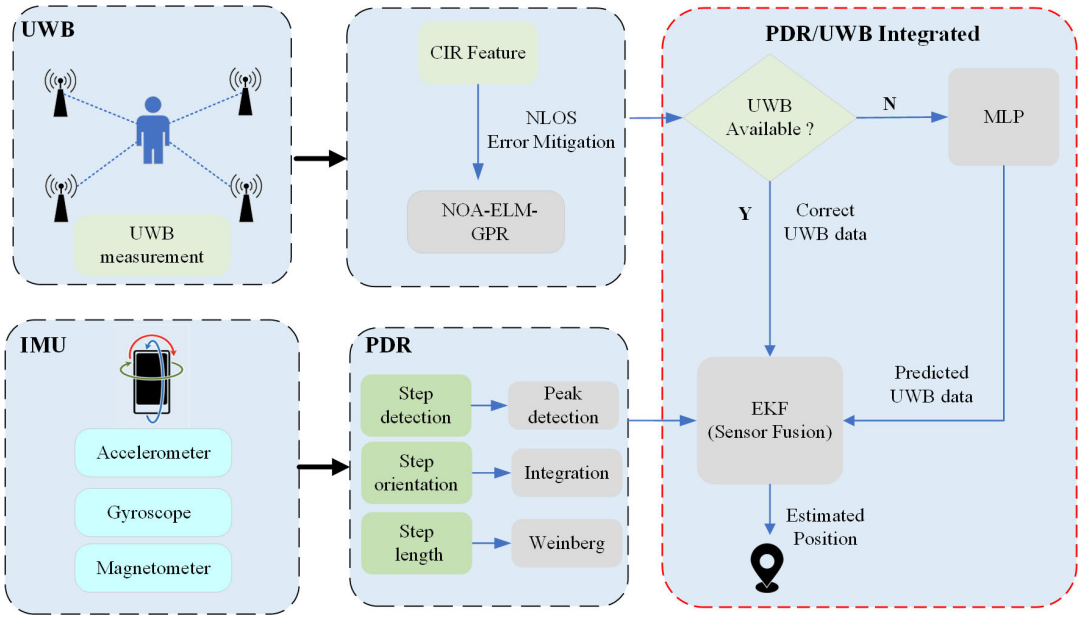


FIGURE 1. Overview of the fused PDR and UWB localization architecture.

TABLE 1. Classification performance under various Z-score thresholding strategies for outlier removal.

Z-score (σ)	Accuracy	Sensitivity (Recall)	Positive Precision	F_1 Metric
2.0	0.936	0.943	0.915	0.928
2.5	0.954	0.950	0.935	0.942
3.0	**0.968**	**0.962**	**0.965**	**0.963**
3.5	0.960	0.950	0.958	0.954
4.0	0.952	0.947	0.953	0.950

To filter out anomalous CIR samples, we adopt a statistical approach based on the Z-score. This method evaluates the degree of deviation of each data point from the dataset mean, measured in units of standard deviation [39]. Data points whose absolute Z-score exceeds a predefined threshold are flagged as outliers and excluded from further analysis.

To assess the influence of Z-score thresholds on the performance of NLOS detection and positioning, we designed an ablation experiment. Results summarized in Table 1 reveal that a threshold value of 3 achieves a favorable trade-off between eliminating abnormal values and retaining meaningful data, thus supporting enhanced classification accuracy and positioning stability.

Once the outliers have been removed, the subsequent phase involves extracting key characteristic parameters derived from the CIR to facilitate NLOS identification and mitigate ranging errors. CIR represents the sum of all received pulses, which can be mathematically expressed as:

$$r(t) = \sum_{l=1}^{L} \beta_l \delta(t - \tau_l) + n(t) \tag{1}$$

where β_l and τ_l correspond to the attenuation coefficient and propagation delay associated with the l-th multipath component, respectively. The variable t denotes the timestamp of each sample within the CIR, which typically comprises 1016 discrete values, each spaced at intervals of 10^{-9} seconds. The additive noise term $n(t)$ is modeled as a Gaussian random process.

Under LOS conditions, the first path in the CIR is usually the most dominant, as the signal encounters minimal obstruction. In contrast, under NLOS conditions, the first path is often attenuated and may not be the strongest. Additionally, the delay spread exhibits significant differences including LOS/NLOS scenarios, providing a useful basis for NLOS identification.

By extracting features such as the amplitude of the first path (Fp_amp1) and the overall CIR amplitude [40], we can effectively identify NLOS conditions. These extracted features serve as inputs to an ELM, which is used for both NLOS classification and ranging error mitigation. This approach not only differentiates between LOS and NLOS conditions but also improves positioning accuracy and reduces ranging errors.

Table 2 provides the formulas used to calculate the key indicators of the CIR, which reflect the characteristics of the communication channel. In this study, we utilized the TDOA (Time Difference of Arrival) method [41] to determine the position of the test point relative to the anchor points. We selected data from five base stations and employed trilateration to determine the unknown location based on geometric principles involving circular and spherical intersections. Specifically, for the distance d_i between each base station located at (x_i, y_i) and the target at (x, y), the following

TABLE 2. Quantitative descriptors for UWB channel impulse response characterization.

Parameter	Technical Specification
Rx_pwr	Composite signal strength metric integrating multipath energy and preamble accumulation statistics
Fp_amp2	Peak amplitude at the second temporal sampling bin following initial path detection
Std_noise	Background noise dispersion (σ), derived from the CIR accumulator
CIR_pwr	Total signal energy normalized by 2^{17} for 16-bit register compatibility
fpindex	Register index corresponding to the detected first path arrival
Fp_pwr	Estimated power of the first path, calculated as $\sum_{k=1}^{3} A_k^2 \cdot \alpha_{\text{PRF}}$, where A_k are component amplitudes and α_{PRF} is the PRF-dependent scaling factor
Rxpacc	Count of successfully accumulated preamble symbols during synchronization
Estimated Range	Estimated distance computed via time-of-flight: $\hat{d}_{ij} = c \cdot \Delta t_{ij}$, where $c = 2.9979 \times 10^8$ m/s and Δt_{ij} is the signal propagation delay
Fp_amp3	Amplitude at the third temporal component after initial path detection
Fp_amp1	Signal magnitude at the first sampling bin post first-path detection

equations can be obtained:

$$\begin{aligned}(x_0 - x)^2 + (y_0 - y)^2 &= d_0^2,\\ (x_1 - x)^2 + (y_1 - y)^2 &= d_1^2,\\ &\cdots\\ (x_n - x)^2 + (y_n - y)^2 &= d_n^2.\end{aligned} \tag{2}$$

These equations describe the position of the target at the intersection of multiple circles determined by the base stations. However, directly solving these nonlinear equations is complex, so we employed the least squares (LS) method [42] to linearize the problem, thereby simplifying the solution process.

To begin with, the matrices a and b are introduced and defined as follows.

$$a = \begin{bmatrix} x_1 - x_0 & y_1 - y_0 \\ x_2 - x_0 & y_2 - y_0 \\ \vdots & \vdots \\ x_n - x_0 & y_n - y_0 \end{bmatrix}, \tag{3}$$

$$\xi = \begin{bmatrix} x \\ y \end{bmatrix}, \tag{4}$$

$$b = \frac{1}{2} \begin{bmatrix} x_1^2 + y_1^2 - d_1^2 - (x_0^2 + y_0^2 - d_0^2) \\ x_2^2 + y_2^2 - d_2^2 - (x_0^2 + y_0^2 - d_0^2) \\ \vdots \\ x_n^2 + y_n^2 - d_n^2 - (x_0^2 + y_0^2 - d_0^2) \end{bmatrix}. \tag{5}$$

The matrix a consists of the positional differences between the base stations, while the vector b combines the inter-station distances with the measured distances. Finally, by solving the linear equation $\xi = (a^T a)^{-1} a^T b$, we can estimate the target's position.

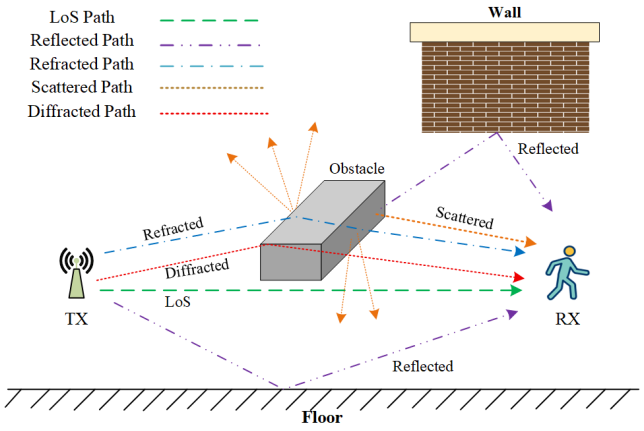


FIGURE 2. Illustration of transmission characteristics under LOS and NLOS conditions.

It is noteworthy that the error introduced under NLOS conditions is generally more substantial than the system noise, which is commonly assumed to follow a Gaussian distribution. As a result, the measured distance tends to exceed the actual value even in LOS scenarios, as illustrated in Fig. 2. When such NLOS-affected measurements are incorporated into positioning algorithms that depend on the spatial geometry of UWB devices, the localization precision can degrade significantly. Consequently, to ensure accurate and real-time positioning in environments containing both LOS and NLOS signals–especially in dynamic contexts–it is essential to employ robust algorithms capable of mitigating NLOS-induced errors.

This process demonstrates how UWB signals and base station positions can be leveraged to estimate the target position using geometric and linear algebraic methods. The application of the least squares method simplifies the solution of complex nonlinear equations. The resulting position data, obtained solely from UWB measurements, will be used as a benchmark for comparison with the proposed method in subsequent sections.

B. PEDESTRIAN DEAD RECKONING (PDR) MODEL

PDR is an autonomous tracking technique that estimates a pedestrian's trajectory by incrementally updating their position on a step-by-step basis. The fundamental principle of PDR is to calculate the position at the current step k based on the position at the previous step $k-1$, the estimated step length L_k, and the heading angle θ_k. This process is governed by the following update equations:

$$\begin{aligned} x_k &= x_{k-1} + L_k \cdot \cos(\theta_k) \\ y_k &= y_{k-1} + L_k \cdot \sin(\theta_k) \end{aligned} \tag{6}$$

where (x_k, y_k) are the coordinates at step k. The accuracy of the PDR system is therefore critically dependent on three core components: gait detection, step length estimation, and heading estimation, which are detailed below.

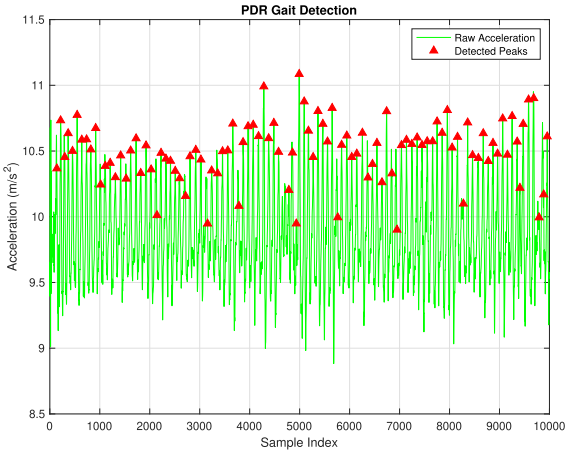


FIGURE 3. Step detection using peak detection method for PDR.

1) GAIT DETECTION

Gait detection is fundamental for accurately identifying each step, which serves as the trigger for position updates [43]. In this work, we employ the Peak Detection Method, which identifies steps by detecting peaks in the accelerometer signal that exceed a certain magnitude and are separated by a minimum time interval. A step is considered detected if the following condition is met:

$$PeakFlag = \begin{cases} 1, & \text{if } \Delta t_p \geq \Delta t_{threshold} \text{ and } a_p \geq \Delta a_p \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

Here, Δt_p is the time difference between consecutive peaks, and a_p is the acceleration magnitude of the current peak. To enhance adaptability, a dynamic acceleration threshold Δa_p is used, which is calculated based on the average acceleration of the previous step, as shown in Eq. (8).

$$\Delta a_p = \begin{cases} \dfrac{1}{\Delta t} \displaystyle\int_{t_{k-1}}^{t_k} a(t)\, dt, & \text{if } k = 2 \\ 1, & \text{if } k = 1 \end{cases} \tag{8}$$

where t_{k-1} and t_k represent the observation times for the beginning and end of $k-1$ and k, and $a(t)$ is the real-time measurement value of the accelerometer. The principle of identifying steps by detecting peaks in the acceleration signal is illustrated in Fig. 3.

2) STEP LENGTH ESTIMATION

Once a step is detected, its length must be estimated. We utilize an improved algorithm based on the Weinberg model [44], which dynamically adjusts the step length by combining historical step data with real-time acceleration characteristics. This approach addresses the limitations of a fixed-parameter model and improves accuracy across varying motion states (e.g., walking, running) and individual gaits. The step length L_k is calculated as:

$$L_k = \alpha \cdot \frac{L_{k-3} + L_{k-2} + L_{k-1}}{3} + (1-\alpha) \cdot \beta \cdot [\max(A_{\text{mode}}) - \min(A_{\text{mode}})]^{1/4} \tag{9}$$

where α and β are control coefficients, and A_{mode} is the modal value of the accelerometer's magnitude during the current step.

3) HEADING ESTIMATION

The heading angle θ_k, which determines the direction of displacement, is obtained by integrating the angular velocity measured by the gyroscope. The yaw angle θ_k^{yaw}, representing the heading in the horizontal plane, is updated at each time sample as follows:

$$\theta_k^{yaw} = \theta_{k-1}^{yaw} + \omega_k^{yaw} \cdot \Delta t \tag{10}$$

where ω_k^{yaw} is the measured angular velocity around the vertical axis at time k, and Δt is the sampling interval. The final heading for a given step is typically determined by averaging the yaw angles over the duration of that step. It is important to note that this integration process is highly susceptible to drift from gyroscope bias, which is a primary source of error in standalone PDR systems.

C. NOA-ELM-GPR ALGORITHM

In this study, we propose a novel NOA-ELM-GPR algorithm to address the NLOS issue in UWB positioning systems and mitigate the ranging errors associated with it. This algorithm integrates the strengths of NOA, GPR, and ELM, aiming to enhance positioning accuracy through optimization and nonlinear modeling. The NOA is a novel nature-inspired metaheuristic algorithm designed to address complex global optimization problems [45]. By mimicking the foraging and storage behaviors of certain animals, NOA effectively explores and exploits the search space, offering a significant improvement over traditional algorithms such as PSO [46] and GWO [47]. NOA achieves a superior balance between exploration and exploitation through its adaptive parameter control and the incorporation of the Lévy flight mechanism.

One of the standout features of NOA is its ability to dynamically switch between different search strategies, making it particularly effective in navigating high-dimensional and complex optimization landscapes. This adaptability reduces the risk of premature convergence, a common issue in many optimization algorithms, thereby enhancing the likelihood of finding the global optimum. Additionally, NOA's fast convergence speed is advantageous in scenarios where time and computational resources are constrained.

The robustness of the algorithm is further attributed to its capability to adaptively regulate parameters in response to the dynamic nature of the search space. This real-time adjustment mechanism allows the algorithm to modify its behavior according to problem-specific characteristics. Additionally, the incorporation of the Lévy flight strategy—which facilitates wide-range, non-local transitions—enables

NOA to avoid premature convergence and to explore the search space more thoroughly. Overall, NOA offers a notable improvement over conventional optimization techniques such as PSO and GWO, delivering greater efficiency and accuracy when tackling complex, high-dimensional global optimization tasks.

Firstly, NOA is utilized for tuning the hyperparameters of the ELM model, such as the number of neurons, activation function selection, and regularization parameters. NOA mimics the behavior of nutcrackers during foraging and food storage, implementing an optimization process divided into exploration and exploitation phases. The core operation of NOA is the position update, which follows different formulas depending on the situation:

$$\mathbf{X}_i^{t+1} = \begin{cases} \mathbf{X}_i^t + \mu \cdot (\mathbf{X}_{\text{best}}^t - \mathbf{X}_i^t) \cdot |\lambda| + r_1 \cdot (\mathbf{X}_A^t - \mathbf{X}_B^t), & \text{if } \tau_1 < \tau_2 \\ \mathbf{X}_{\text{best}}^t + \mu \cdot (\mathbf{X}_A^t - \mathbf{X}_B^t), & \text{if } \tau_1 < \tau_3 \\ \mathbf{X}_{\text{best}}^t \cdot l, & \text{otherwise} \end{cases} \tag{11}$$

where $\mathbf{X}_i^t$ denotes the location of the i-th agent during the t-th iteration, and $\mathbf{X}_{best}^t$ represents the optimal solution identified at the same iteration, λ is a random number based on Lévy flight to control step size randomness, and r_1 is a random number between 0 and 1. $\mathbf{X}_A^t$ and $\mathbf{X}_B^t$ are the positions of two randomly selected individuals, and l is a factor that decreases linearly from 1 to 0 to gradually reduce the search space.

In different phases of the algorithm, NOA dynamically adjusts the search strategy between exploration and exploitation. During the exploration phase, the position update rule is typically:

$$\mathbf{X}_i^{t+1} = \mathbf{X}_i^t + \mu \cdot (\mathbf{X}_A^t - \mathbf{X}_B^t) + \mu \cdot (r_2 \cdot U_j - L_j) \tag{12}$$

In the exploitation phase, the update formula may become:

$$\mathbf{X}_i^{t+1} = \mathbf{X}_C^t + \mu \cdot (\mathbf{X}_A^t - \mathbf{X}_B^t) + \mu \cdot (r_1 \cdot U_j - L_j) \tag{13}$$

where U_j and L_j are the upper and lower bounds of the search space, r_2 and r_1 are random numbers between 0 and 1, and $\mathbf{X}_C^t$ is a candidate solution found by the algorithm.

Additionally, as the algorithm approaches the optimal solution, NOA incorporates a cache-search and recovery mechanism to optimize the search process further. Specifically, when optimization reaches a certain stage, individuals may "remember" previously found high-quality solutions and attempt to recover them:

$$\mathbf{X}_i^{t+1} = \begin{cases} \mathbf{X}_i^t, & \text{if } f(\mathbf{X}_i^t) < f(\text{RP}_{i,1}^t) \\ \text{RP}_{i,1}^t, & \text{otherwise} \end{cases} \tag{14}$$

where $RP_{i,1}^t$ is the cached reference point location. If the current position is better than the cached reference point, the current position is retained; otherwise, it is updated to the cached reference point. The main steps are summarized in Algorithm 1 as follows.

Initially proposed for supervised learning tasks, the ELM has seen its application scope expand significantly in subsequent research [48]. Due to its ability to handle both classification and regression problems, ELM has garnered considerable attention from experts and researchers. The optimization process of ELM enables it to dynamically adjust its structure to fit various datasets and environmental conditions, thus improving overall performance.

In the context of UWB positioning, ELM proves highly efficient for modeling positioning data. As a feedforward neural network with a single hidden layer, the ELM is

Algorithm 1 NOA for ELM Parameter Optimization

Input: y, lb, ub, D, N, T
Output: *Best_pos*, *Best_score*, *curve*

Initialize parameters: $\delta = 0.05$, $Prp = 0.2$, $Pa2 = 0.4$
Initialize population *pop* with random positions within *lb* and *ub*
Calculate cost for each individual in *pop*
Determine the best solution *X_best*
for $it = 1$ to T **do** ▷ Main Loop
  Calculate mean position *X_mean*
  Update $Pa1 = 1 - it/T$ and $l = 1 - it/T$
  Update alpha using Eq. (11)
  if rand() > rand() **then** ▷ Foraging and Storage Strategy
    for each individual i in *pop* **do**
      Calculate u based on $r1$, $r2$, $r3$
      if rand() > $Pa1$ **then**
        Update *newpop*[i].*Position* using Eq. (4)
      else
        Exploit *newpop*[i].*Position* using Eq. (3)
      end if
      Enforce boundary constraints
      Evaluate new cost and perform greedy selection
    end for
  else ▷ Cache-Search and Recovery Strategy
    for each individual i in *pop* **do**
      Update reference points $PR1[i]$ and $PR2[i]$
    end for
    for each individual i in *pop* **do**
      if *phi* > $Pa2$ **then** ▷ Exploration Phase 2
        Update *newpop*[i].*Position* using Eq. (16)
      else ▷ Exploitation Phase 2
        Select best among *pop*[i], $PR1[i]$, $PR2[i]$
      end if
      Enforce boundary constraints
      Evaluate new cost and perform greedy selection
    end for
  end if
  Sort *pop* by cost and update *X_best*
  Record best cost *BestCost*[it]
end for
return *Best_pos*, *Best_score*, *curve*

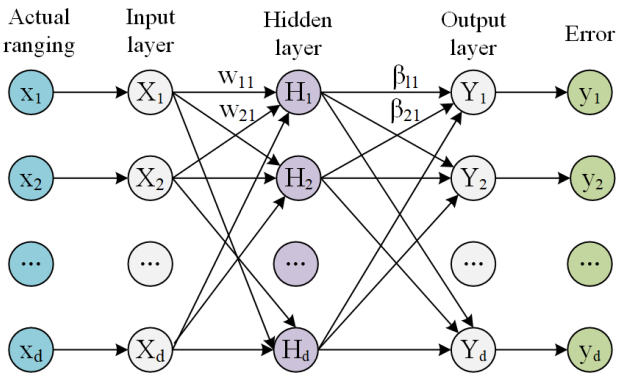


FIGURE 4. ELM Framework for Error Compensation in Ranging.

characterized by the random assignment of input-to-hidden layer weights and biases. The output weights are subsequently computed by solving a linear system using the least squares method. This architecture provides notable advantages, including faster training speeds and enhanced generalization capabilities [49]. Beyond its predictive accuracy in reducing positioning errors, ELM can also detect potential signal interference during the UWB-based mobile robot positioning process, further contributing to its robustness.

In this work, an ELM-based model is constructed to learn the nonlinear mapping between the ranging measurements (obtained from the target-anchor pairs) and their corresponding categories or error values, as illustrated in Fig. 4.

Suppose there are N groups of training samples for ranging between six anchors $(a_0, a_1, a_2, a_3, a_4, a_5, a_6)$ and the target, represented as $(x_i, t_i) \in \mathcal{R}^n \times \mathcal{R}^m$. The ELM model can thus be expressed as:

$$y_j = \sum_{i=1}^{l} \beta_{ij} G_i(\omega_i, b_i, X), \quad j = 1, 2, \ldots, N \tag{15}$$

where:

- β_{ij} represents the weight connecting the i-th hidden node to the output layer,
- ω_i is the weight vector connecting the i-th hidden node to the input layer,
- b_i is the bias of the i-th hidden node,
- $G_i(\omega_i, b_i, X)$ is the activation function output, typically represented by a Sigmoid function:

$$G_i(\omega_i, b_i, X) = \frac{1}{1 + \exp\left(-(\omega_i \cdot X + b_i)\right)} \tag{16}$$

The hidden layer's output matrix H is formulated as follows:

$$H = \begin{bmatrix} G_1(\omega_1 \cdot x_1 + b_1) & \ldots & G_l(\omega_l \cdot x_1 + b_l) \\ G_1(\omega_1 \cdot x_2 + b_1) & \ldots & G_l(\omega_l \cdot x_2 + b_l) \\ \vdots & \ddots & \vdots \\ G_1(\omega_1 \cdot x_Q + b_1) & \ldots & G_l(\omega_l \cdot x_Q + b_l) \end{bmatrix} \tag{17}$$

The output weight matrix β is then determined using the least squares method:

$$\beta = H^+ T \tag{18}$$

where H^+ denotes the Moore–Penrose pseudoinverse of the matrix H, and T represents the desired output matrix. This approach enables ELM to perform fast training and prediction on large datasets.

Finally, the results of ω, β, b are substituted into equation (9) to obtain the output layer results.

Next, GPR is applied to correct the errors in the ELM output. GPR measures the similarity between input data points using a kernel function and provides nonlinear predictions based on these similarities. The prediction equation for GPR is:

$$\hat{y} = k(x, X) \cdot [K(X, X) + \sigma_n^2 I]^{-1} \cdot Y \tag{19}$$

where:

- $k(x, X)$ is the kernel matrix between the input point x and the training set X,
- $K(X, X)$ is the autocorrelation matrix of the training dataset,
- σ_n^2 is the noise variance.

The Matern kernel, often used in GPR for modeling nonlinear errors in NLOS environments, is expressed as:

$$K(x, x') = \sigma^2 \left(1 + \frac{\sqrt{5}d}{l} + \frac{5d^2}{3l^2}\right) \exp\left(-\frac{\sqrt{5}d}{l}\right) \tag{20}$$

where d is the distance between input data points, l is the length scale parameter, and σ is the signal variance. GPR thus provides an accurate correction to the ELM output, further reducing the positioning error. Algorithm 2 presents the full sequence of steps involved in the implementation of the proposed method.

D. FUSION OF EKF-MLP MODEL

The fusion of MLP and EKF aims to improve the accuracy of position estimation, particularly in environments characterized by nonlinear dynamics and noise. This approach leverages the strong nonlinear fitting capability of MLP and the state estimation capability of EKF to achieve more precise system state estimation.

In the fusion process, EKF [50] is first used to predict the state. EKF predicts the state vector and covariance matrix at the next time step based on the system's motion model:

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1}, u_{k-1}) \tag{21}$$

$$P_{k|k-1} = \Phi_{k-1} P_{k-1} \Phi_{k-1}^T + Q_k \tag{22}$$

where $\hat{x}_{k|k-1}$ is the predicted state vector, $P_{k|k-1}$ is the predicted covariance matrix, Φ_{k-1} is the state transition matrix, and Q_k is the process noise covariance matrix.

The MLP, a representative architecture within the class of feedforward neural networks, has been extensively utilized across various learning tasks. Its architecture consists of

Algorithm 2 NOA-ELM-GPR for NLOS Error Mitigation

Input: Training dataset P_{train}, T_{train}; Validation dataset P_{verfy}, T_{verfy}; Test dataset P_{test}, T_{test};
Output: Corrected ranging error using NOA-ELM-GPR

Step 1: Data Preparation
Load and extract features from all datasets
Step 2: NOA for ELM Parameter Optimization
Set number of neurons $N \leftarrow 200$
Define objective function $f_1 \leftarrow fun(x, P_{train}, T_{train}, P_{verfy}, T_{verfy}, N)$
Set NOA parameters: dimension $D \leftarrow 6 \times N$, population $pop \leftarrow 80$, max iterations $MaxIt \leftarrow 50$, bounds $lb, ub \leftarrow [-100, 100]$
Execute NOA to optimize ELM parameters: $[X, Best_score, curve] \leftarrow NOA(f_1, lb, ub, D, pop, MaxIt)$
Step 3: ELM Model Training and Testing
Train ELM model using optimized parameters on P_{train}, T_{train}
Validate model: $T_{sim_verfy} \leftarrow elmpredict(P_{verfy}, X)$
Test model: $T_{sim_test} \leftarrow elmpredict(P_{test}, X)$
Step 4: GPR for Error Correction
Compute Matern kernel for GPR and predict signal strength:
for $i = 1$ to len_tp **do**
$test_rp_rss(i) \leftarrow k_t \cdot K_{cov}^{-1} \cdot (T_{verfy} - T_{sim_verfy})' + T_{sim_test}(i)$
end for
Step 5: Error Calculation and Result Analysis
Compute error metrics for ELM, GPR, and raw data
Analyze performance of error compensation methods
Step 6: Plot Results
Plot convergence curve and compare error mitigation methods

an input layer, several hidden layers, and an output layer positioned at the end, enabling it to handle both classification and regression problems effectively. Owing to its capacity for modeling intricate nonlinear relationships, MLP is frequently adopted as a foundational learner in stacked generalization schemes to improve ensemble prediction accuracy [51].

In the context of this study, the MLP neural network [52] is utilized to correct the initial predictions of the EKF by learning the system's nonlinear error characteristics. The structure of the MLP, as illustrated in Fig. 5, includes 7-dimensional input features and a 2-dimensional output (X, Y coordinates). The network comprises 50 hidden units in its hidden layer. As a typical feedforward neural network, the MLP processes raw input data through weighted summation and bias adjustments before transmitting it to the hidden layer. The hidden layer, which generally contains more neurons than the input layer, applies a nonlinear activation function to introduce nonlinearity into the model. Finally, the output

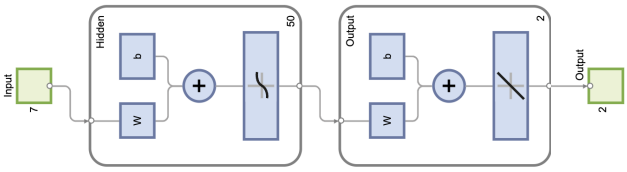


FIGURE 5. The trained MLP with 50 neurons in the hidden layer.

layer aggregates the processed signals from the hidden layer to generate the final prediction.

In our study, we considered a time window of 10 seconds for utilizing MLP predictions during the absence of UWB signals. This choice is based on a balance between prediction accuracy, system stability, and computational cost. While shorter time windows (e.g., 1 to 3 seconds) could theoretically reduce cumulative error, they would require frequent switching between different positioning modes, potentially affecting system stability and increasing computational overhead. The 10-second window was chosen to allow sufficient time for UWB signals to recover, ensuring smoother system transitions without frequent interruptions. Additionally, experimental analysis indicated that, although the prediction error tends to accumulate over time, the level of accuracy achieved within this 10-second period remains acceptable for the intended application scenarios, particularly in less complex environments where fewer obstacles and signal reflections are present. This compromise allows the system to maintain reasonable accuracy while also managing computational efficiency and ensuring robustness in real-time application contexts.

MLP predicts system errors based on historical data and measurements, which are then used to correct the EKF prediction. The training of MLP involves optimizing weights and biases to effectively learn the mapping relationship between input features and output results, making it widely applicable in tasks such as classification and regression.

$$\Delta\hat{x}_k = MLP(Z_k, u_k) \tag{23}$$

where $\Delta\hat{x}_k$ is the state correction term from MLP, and Z_k represents the observation data.

The state update step combines the EKF prediction and MLP correction to produce a more accurate state estimation:

$$\hat{x}_k = \hat{x}_{k|k-1} + K_k(Z_k - H_k\hat{x}_{k|k-1}) + \Delta\hat{x}_k \tag{24}$$

where K_k is the Kalman gain, H_k is the observation matrix, and Z_k is the measurement value.

Finally, the covariance matrix is updated using the standard EKF update equation, further refining the uncertainty estimation of the state.

$$P_k = (I - K_k H_k)P_{k|k-1} \tag{25}$$

This fusion method combines the filtering capabilities of EKF with the nonlinear mapping capabilities of MLP, significantly enhancing the accuracy of state estimation in

Algorithm 3 PDR/UWB Fusion for Pedestrian Localization

Input: Initial settings *sig_size*, *acc_thres*
Output: Estimated pedestrian location $\hat{L}$

Initialization:
Set signal and acceleration thresholds
Step 1: Import Data
Load test positions, UWB base station data, and distance measurements
Step 2: Initial Location Estimation
Estimate initial location using UWB and TDOA
Step 3: IMU Data Processing
Load IMU data and estimate heading
Detect steps and calculate step lengths
Step 4: EKF Location Estimation
Apply EKF to fuse IMU and UWB data for location estimation
Step 5: MLP Estimation for UWB Loss (81s to 90s)
Train MLP with data from the first 80 seconds
Predict locations for 81-90 seconds using MLP
for $i = 81$ to 90 **do**
    Set $ini_loc_mlp(i, :) \leftarrow mlp_loc(i - 80, :)$ ▷ Update location using MLP predictions
end for
Step 6: EKF with MLP Observations
Apply EKF with updated MLP-based observations for final location
Step 7: Error Analysis
Calculate RMSE and plot the results

complex dynamic systems. Algorithm 3 outlines the step-by-step pseudocode for realizing the proposed method.

E. COMPUTATIONAL COMPLEXITY ANALYSIS

The computational complexity of our framework consists of two distinct stages: offline optimization and online real-time localization.

The offline stage is dominated by NOA–ELM optimization and GPR training. NOA–ELM optimization has a complexity of $O(T_{\max}N_{\text{pop}}L^3)$, with $T_{\max}$ iterations, population size N_{pop}, and L hidden neurons. GPR training complexity is $O(Q^3)$, mainly due to covariance matrix inversion for Q training points. Practical complexities are significantly reduced through parallel computing and early stopping, as reflected in our MATLAB implementation.

The online stage involves three components per UWB update: the PDR update ($O(1)$), NLOS mitigation–consisting of ELM inference ($O(D_{\text{in}}L)$) and optimized GPR inference ($O(Q)$), and EKF–MLP fusion, which includes EKF updates ($O(n^3)$, $n = 3$) and occasional MLP inference ($O(D_{\text{in}}H + HD_{\text{out}})$). Table 3 summarizes these streamlined complexities.

On an Intel Xeon 8358P server (2.60 GHz), the complete offline training stage finishes in approximately 8 minutes, while the online inference latency consistently remains below 5 ms per frame, thereby satisfying the real-time requirement of a 10 Hz UWB update rate.

TABLE 3. Summary of Dominant Complexities.

Component	Time Complexity	Space Complexity
NOA–ELM optimization	$O(T_{\max}N_{\text{pop}}L^3)$	$O(N_{\text{pop}} \times d)$
GPR training	$O(Q^3)$	$O(Q^2)$
ELM inference	$O(D_{\text{in}}L)$	$O(L)$
GPR inference (optimized)	$O(Q)$	$O(Q)$
PDR update	$O(1)$	$O(1)$
EKF update	$O(n^3)$	$O(n^2)$
MLP inference	$O(D_{\text{in}}H + HD_{\text{out}})$	$O(H)$

IV. EVALUATION AND ANALYSIS

In this section, we first validate the effectiveness of the proposed algorithm in mitigating UWB ranging errors using public datasets. Subsequently, we evaluate the experimental results of the proposed algorithm through dedicated experimental setups to demonstrate its applicability and superiority in indoor environments.

A. SIMULATION VERTIFICATION

To validate the effectiveness of our proposed NLOS mitigation algorithm in a controlled and reproducible environment, we utilized a publicly available dataset from the Institute of Techno-Informatics at Graz University of Technology, Austria [53]. The dataset was collected using Decawave's DWM1001-DEV device across three indoor scenarios: an office, a laboratory, and a corridor, comprising a total of 36,700 samples with both LOS and NLOS conditions. For our evaluation, we performed 10-fold cross-validation, partitioning the data into a 70% training set and a 30% testing set. In this process, the ELM model was trained on the training split to learn the mapping from the extracted UWB CIR features to two corresponding ground-truth labels: a binary class indicating the channel state (LOS/NLOS) and the true ranging error. To optimize computational resources, CIR data was processed starting from the 700th sample onward.

A robust mitigation framework fundamentally relies on the accurate classification of LOS and NLOS signals. This classification step directly determines if a UWB measurement is reliable or "available" for direct use; a signal classified as NLOS is considered "unavailable" for raw positioning and requires mitigation. Therefore, before evaluating the final ranging error, we first assessed the performance of our core NOA-ELM identification model against baseline SVM and CNN classifiers. The results, averaged across the 10-fold cross-validation, are summarized in the following table.

As shown in Table 4, our proposed NOA-ELM model achieves superior performance across all classification metrics. It obtained an accuracy of 96.8%, outperforming both SVM and CNN. More critically, its high F1-Score of 96.3% indicates an excellent balance between Precision and Recall,

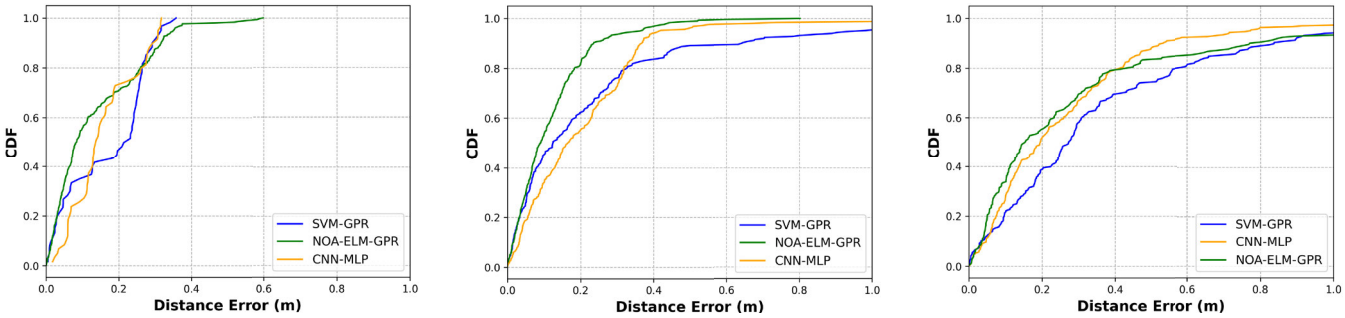


FIGURE 6. Comparative analysis of NLOS error mitigation. Each subfigure shows the CDF distribution of ranging errors under (a) laboratory, (b) office, and (c) corridor scenarios.

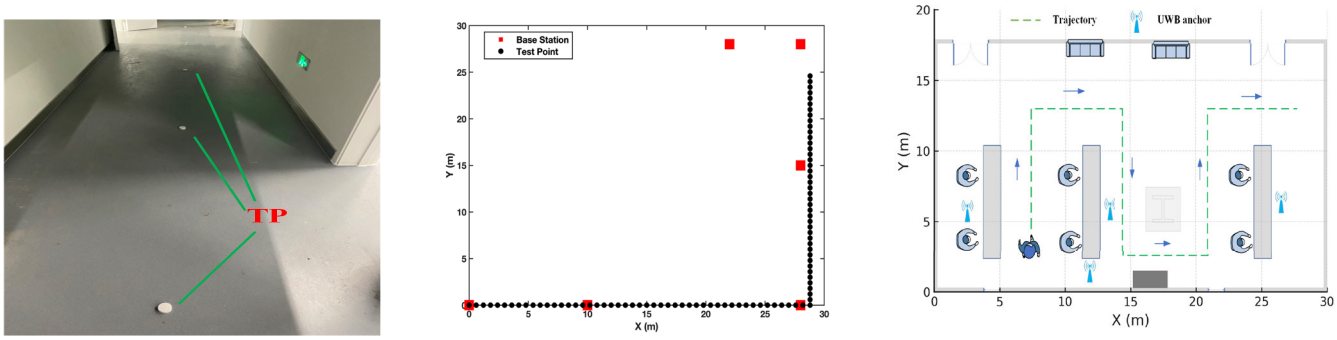


FIGURE 7. Setup for UWB/PDR Fusion Trajectory Testing. (a) Corridor Test Points. (b) Corridor Trajectory Layout. (c) Office Trajectory Layout.

confirming its reliability in correctly identifying NLOS conditions while avoiding the misclassification of LOS signals. Furthermore, the NOA-ELM model is significantly more efficient, with an inference time of just 1.2 ms per sample. This high-fidelity and efficient classification provides a strong foundation for the subsequent error mitigation stage.

Building upon this superior NLOS identification capability, we then evaluated the end-to-end performance of the complete mitigation algorithms: SVM-GPR, a CNN-based model, and our proposed NOA-ELM-GPR. The CDF of the final distance error was analyzed across the three environments, as shown in Fig. 6. In all scenarios–laboratory, office, and corridor–the proposed NOA-ELM-GPR algorithm consistently outperformed the baseline methods, achieving a higher probability of lower ranging errors.

The quantitative results in Table 5 further substantiate these findings. The NOA-ELM-GPR algorithm demonstrated the lowest Root Mean Square Error (RMSE) and standard deviation in all three environments, highlighting its effectiveness and stability. Overall, these experimental results confirm that the superior classification accuracy of the NOA-ELM module directly translates into more effective ranging error mitigation by the complete NOA-ELM-GPR model. This enhanced accuracy in distance measurement is crucial for the subsequent UWB/PDR fusion, ensuring more reliable indoor pedestrian tracking.

TABLE 4. Performance and Efficiency Comparison of NLOS Identification Models.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Inference Time (ms/sample)
SVM	92.5	91.8	92.1	91.9	2.4
CNN	94.2	93.6	93.9	93.7	8.6
Ours	**96.8**	**96.5**	**96.2**	**96.3**	**1.2**

Times averaged over 500 runs on Intel Xeon 8358P server (2.60 GHz).

TABLE 5. Distance Error Comparison in Different Environments.

Scenarios	Metric(m)	SVM-GPR	NOA-ELM-GPR	CNN-MLP
Lab	RMSE	0.20	0.15	0.19
	STD	0.11	0.08	0.12
	Mean	0.14	0.10	0.13
Office	RMSE	0.33	0.18	0.26
	STD	0.25	0.12	0.20
	Mean	0.22	0.14	0.19
Corridor	RMSE	0.47	0.41	0.50
	STD	0.30	0.25	0.37
	Mean	0.35	0.20	0.27

B. EXPERIMENTAL SETUP

To evaluate the end-to-end performance of the proposed fusion framework in a real-world dynamic scenario, we

TABLE 6. Positioning Errors in Corridor and Office Scenarios.

Methods	Corridor Scenario (m)				Office Scenario (m)			
	Max Error	Min Error	Avg Error	RMSE	Max Error	Min Error	Avg Error	RMSE
PDR	1.84	0.33	1.32	1.54	2.56	0.47	1.52	1.74
UWB	1.53	0.15	0.84	1.02	1.72	0.23	0.92	1.13
EKF	1.02	0.22	0.62	0.73	1.23	0.34	0.62	0.83
Ours	**0.82**	**0.02**	**0.32**	**0.40**	**1.02**	**0.04**	**0.53**	**0.62**

conducted a pedestrian tracking experiment in a corridor and office environment within the School of Electrical and Electronic Engineering at Universiti Sains Malaysia (USM). The experimental setup, including the path and UWB base station (BS) deployment, is depicted in Fig. 7.

The primary hardware for absolute positioning was the Decawave DW1000 UWB system, comprising six fixed BSs and one mobile tag. For relative tracking, a consumer-grade smartphone equipped with a triaxial accelerometer, gyroscope, and magnetometer was used to collect IMU data for the PDR algorithm, with a sampling rate of 50 Hz. During the experiment, a researcher held the smartphone and walked along a predefined path, stepping on landmarks with known coordinates to establish ground truth. This process allowed us to collect a comprehensive local dataset containing synchronized UWB CIR data, UWB range measurements, IMU readings, and corresponding ground truth positions. All data was recorded for offline processing.

To address the reviewer's question regarding the acquisition of training samples, we detail the training process for both the ELM and MLP models below. Both models were trained and evaluated using the data collected from our local USM dataset.

For the NOA-ELM model (NLOS Mitigation): The data collected from the USM experimental runs was partitioned to train and test the NLOS mitigation module. We allocated 70% of the collected samples as the training set and the remaining 30% as the testing set. The ELM was trained to learn the mapping from the UWB CIR features (input) to the true ranging error (output). The ground truth ranging error for each sample was calculated by subtracting the true distance (derived from the known landmark coordinates) from the measured UWB range.

For the MLP model (UWB Prediction): The MLP, which is tasked with predicting pseudo-UWB positions during signal outages, was trained in an online, trajectory-specific fashion. For each experimental run, the synchronized PDR data and the corresponding valid UWB position measurements from the first 80 seconds of the trial were used as the training set. The MLP was thus trained to learn the specific mapping from PDR inputs to UWB position outputs for that particular environment. Subsequently, during periods of simulated UWB signal loss, this trained model was used to generate position predictions based only on the live PDR data.

To rigorously evaluate our proposed EKF-MLP algorithm and address the reviewer's request for separate evaluations, we compared its performance against three essential baselines: a standalone PDR system (PDR-only), a standalone UWB system (UWB-only), and a conventional EKF-based fusion algorithm (EKF). The performance of all methods was evaluated against the ground truth using the RMSE metric.

C. EXPERIMENTAL RESULTS

1) NLOS IDENTIFICATION AND RANGE ERROR MITIGATION

The stability and effectiveness of our optimization strategy are first demonstrated by the convergence process of the NOA, depicted in Fig. 8a. The fitness value progressively decreases and stabilizes at a minimum after 38 iterations, indicating that a near-optimal set of parameters for the model has been found. Leveraging these optimized parameters, we evaluated the framework's performance on NLOS mitigation. As shown in Fig. 8b, the raw UWB signals exhibit significant ranging errors in the NLOS environment, with fluctuations reaching up to 6 meters. While a standard ELM model improves accuracy, it still suffers from considerable error volatility. In contrast, our proposed NOA-ELM-GPR algorithm achieves a significantly smoother and more stable error curve. This superior performance in accuracy and reliability is further corroborated by the CDF analysis in Fig. 8c. The CDF curve for the NOA-ELM-GPR algorithm rises much more rapidly than the others, confirming its ability to consistently deliver high-precision ranging results.

To further assess our method, its performance was benchmarked against other algorithms by evaluating the trade-off between ranging error and computational time, as illustrated in Fig. 8d. The results highlight a clear performance spectrum. The standard ELM provides the quickest results (8 s) but with the largest average error (40 cm). Algorithms incorporating parameter optimization, such as PSO-ELM (30 cm error, 14 s) and BO-SVM (25 cm error, 12 s), offer a more balanced compromise between efficiency and accuracy. Notably, our proposed NOA-ELM-GPR algorithm achieves the highest precision with an average ranging error of only 8 cm. This superior accuracy comes at the expense of the longest test time (16 s), positioning it as the ideal solution for applications where high-precision localization is the primary requirement.

2) UWB/PDR FUSION LOCALIZATION

We comprehensively evaluated the proposed EKF-MLP algorithm against several benchmarks, including a standalone PDR system (PDR-only), a standalone UWB system (UWB-only), and a conventional EKF-based fusion approach (EKF). The experiments were conducted in both corridor and office environments, aiming to demonstrate the robustness of the proposed algorithm under different signal conditions and environmental complexities. Overall, our results show that the proposed EKF-MLP consistently outperforms the baseline methods, achieving superior localization accuracy and robustness to NLOS effects.

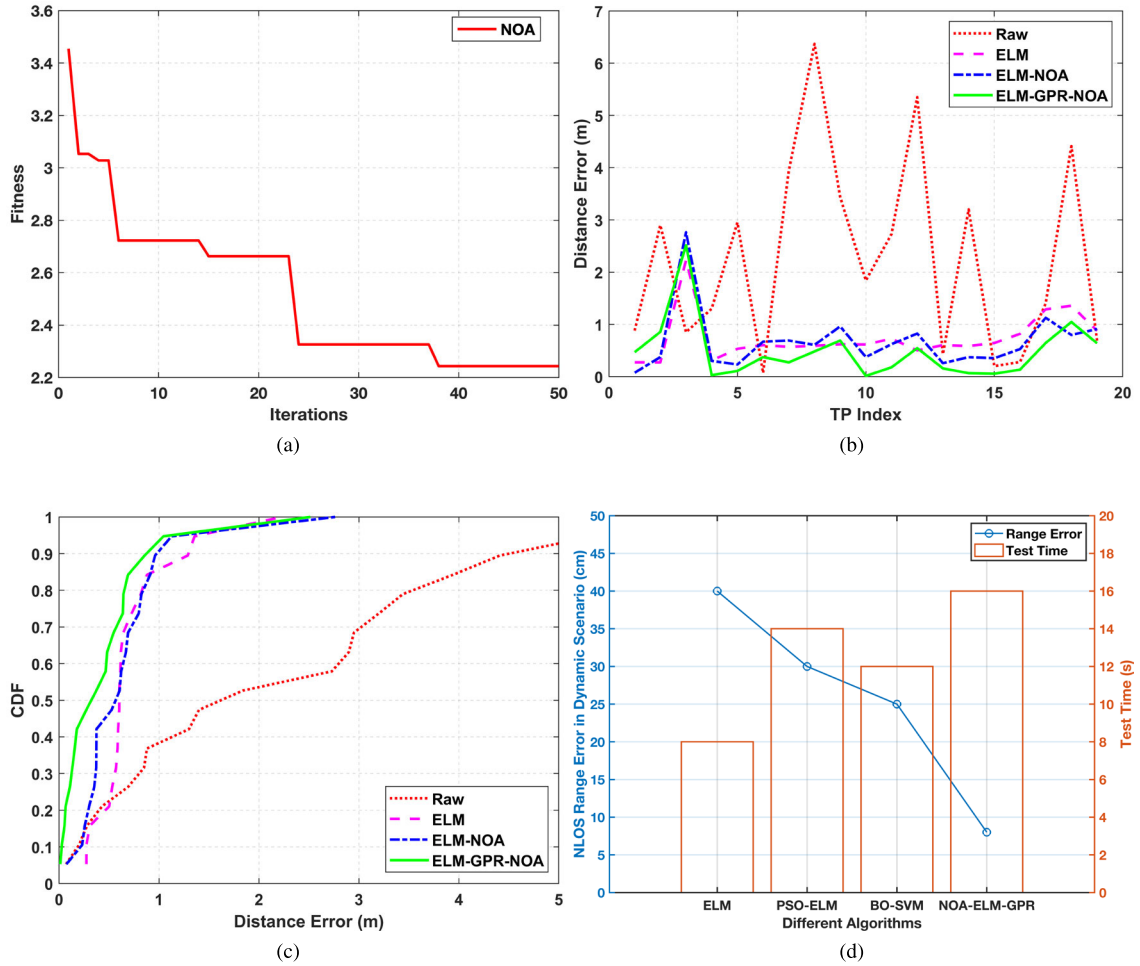


FIGURE 8. Comparative analysis of NLOS error mitigation: (a) Loss value progression with iterations for the NOA algorithm; (b) Localization errors at 19 test points; (c) CDF comparison of four algorithms; (d) Overall performance comparison of different methods.

In the corridor scenario, the PDR-only system exhibited significant trajectory drift and rapid deviation from the ground truth, as shown by the red dashed line in Fig. 9. This is primarily due to the uncorrected heading drift inherent in PDR, leading to error accumulation over time. Both the UWB-only and conventional EKF methods also faced performance degradation as UWB signals became intermittent or unavailable. In contrast, the EKF-MLP algorithm maintained close alignment with the ground truth throughout the test, leveraging its predictive capability to compensate for UWB signal loss. The CDF of the localization errors in Fig. 9b further highlights the substantial improvement achieved by the proposed approach.

In the more challenging office environment, characterized by frequent NLOS effects and severe signal blockage, the advantages of the proposed method became even more pronounced. As illustrated in Fig. 9c, the EKF-MLP trajectory consistently followed the ground truth path, while the EKF-only and UWB-based systems suffered from notable localization degradation. The quantitative results in Fig. 9d confirm that our method effectively mitigates the impact of NLOS errors and maintains high positioning accuracy, even in complex indoor settings.

To further illustrate the performance of each localization method in different environments, Table 6 summarizes the positioning errors across both corridor and office scenarios.

In the corridor scenario, the proposed EKF-MLP algorithm significantly outperformed the other methods. Specifically, the PDR approach resulted in a maximum positioning error of 1.84 meters and an average error of 1.32 meters, with a RMSE of 1.54 meters. Similarly, the UWB-only method produced a maximum error of 1.53 meters, an average error of 0.84 meters, and an RMSE of 1.02 meters. Integrating UWB with PDR using an EKF improved accuracy, reducing the maximum error to 1.02 meters, the average error to 0.62 meters, and achieving an RMSE of 0.73 meters. The proposed EKF-MLP method achieved the lowest positioning errors, with a maximum error of 0.82 meters, a minimum error of 0.02 meters, an average error of 0.32 meters, and an RMSE of 0.40 meters.

In the office scenario, a more challenging environment due to increased NLOS conditions and complex obstacles, the

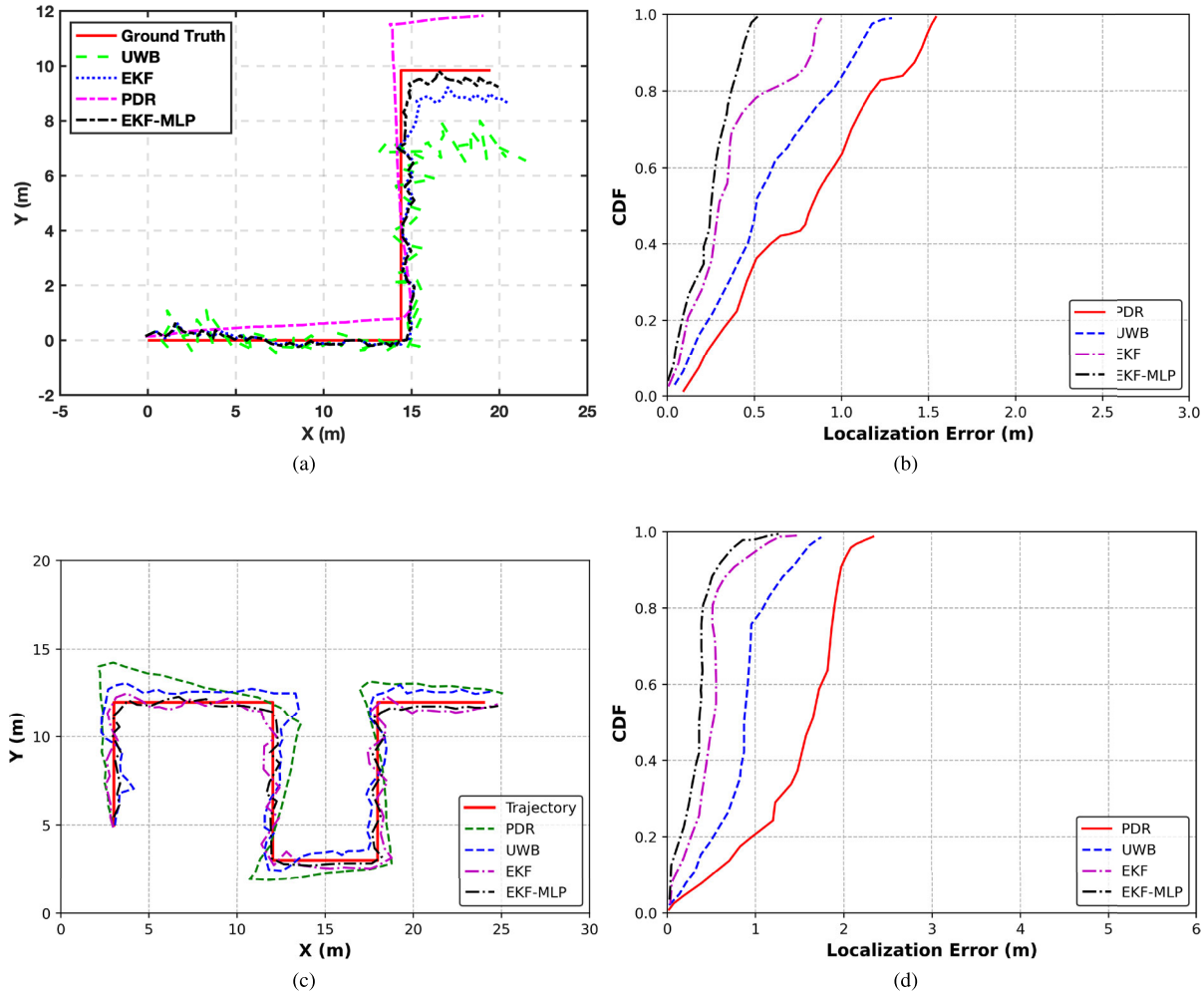


FIGURE 9. Trajectory and Positioning Error Comparison: (a) Corridor Trajectories, (b) Corridor CDF, (c) Office Trajectories, (d) Office CDF.

proposed method continued to demonstrate its robustness. The PDR method showed a maximum error of 2.56 meters and an RMSE of 1.74 meters, while UWB-only resulted in a slightly lower maximum error of 1.72 meters and an RMSE of 1.13 meters. The EKF-based integration of UWB and PDR further reduced these errors, achieving a maximum error of 1.23 meters and an RMSE of 0.83 meters. However, the EKF-MLP method outperformed all other methods by yielding a maximum error of 1.02 meters, a minimum error of 0.04 meters, an average error of 0.53 meters, and the lowest RMSE of 0.62 meters.

These results clearly indicate that the proposed EKF-MLP algorithm provides significant improvements in positioning accuracy across different indoor environments. Compared to standalone methods, the EKF-MLP approach offers substantial reductions in both maximum and average positioning errors as well as RMSE, highlighting its capability to handle diverse scenarios effectively. Importantly, the EKF-MLP method maintains high localization accuracy even in challenging office environments characterized by increased NLOS conditions, making it a robust solution for indoor positioning.

TABLE 7. Comparison of Position Errors for Different Targets.

Target	Method	Avg Error (m)	Max Error (m)	Min Error (m)
1	PDR	1.21	2.23	0.68
	UWB	0.82	1.94	0.14
	Ours	0.55	1.27	0.08
2	PDR	1.15	2.72	0.77
	UWB	0.75	1.65	0.13
	Ours	0.44	1.05	0.07
3	PDR	1.52	2.62	0.64
	UWB	0.73	2.15	0.22
	Ours	0.52	1.24	0.05

To further validate the stability of the proposed system, additional experiments were carried out in an office environment across three different targets. The results consistently indicate that our method achieves superior positioning accuracy compared to the baseline approaches.

As illustrated in Fig. 10, the error distributions for Target 1, Target 2, and Target 3 under all three methods were compared. The proposed approach (“Ours”) exhibits the smallest error ranges across all targets, confirming its adaptability and robustness in various indoor conditions. In contrast, both PDR and UWB methods show larger and more variable error

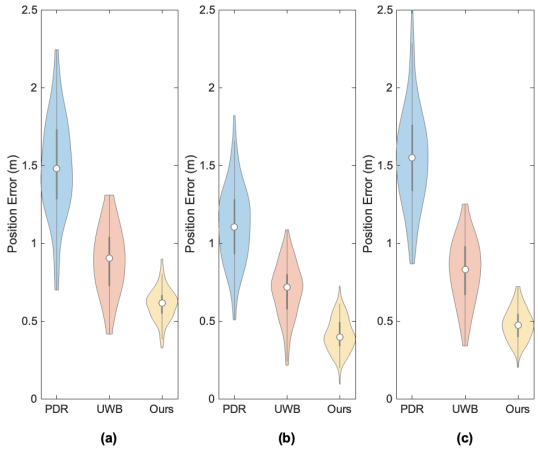


FIGURE 10. Positioning Error Comparison for Different Methods. (a) Target 1, (b) Target 2, (c) Target 3.

distributions, particularly in Target 1 and Target 3, where PDR errors are markedly higher.

Table 7 further details the average, maximum, and minimum errors for each target. For Target 1, PDR yields an average error of 1.21 m and UWB attains 0.82 m, while the proposed method reduces this to only 0.55 m. Similar trends are observed for Target 2 and Target 3, with the proposed approach maintaining consistently lower average errors at 0.44 m and 0.52 m, respectively. Both maximum and minimum error metrics similarly highlight the superior stability of the proposed method compared to PDR and UWB.

In summary, the experimental findings confirm that the proposed EKF-MLP algorithm provides high positioning accuracy and stability under diverse target conditions and complex indoor environments. Compared to standalone PDR or UWB methods, the proposed approach substantially reduces positioning errors, as evidenced by the experimental data. These results strongly validate the feasibility and effectiveness of the proposed method for challenging indoor localization tasks.

V. CONCLUSION

UWB-based indoor positioning methods frequently suffer from low sampling rates and are highly susceptible to NLOS conditions, while PDR methods often produce lower accuracy and encounter error accumulation over time. To address these issues, this paper proposed an enhanced PDR/UWB fusion algorithm. A novel NLOS error mitigation strategy was developed by integrating ELM and GPR, with initial parameters optimized by NOA. In addition, an MLP-based UWB/PDR fusion approach was introduced to handle scenarios where UWB signals are intermittently unavailable, allowing the MLP predictions to serve as PDR observations in the absence of UWB measurements.

The experimental results demonstrated that the proposed NLOS error mitigation technique substantially reduced the impact of NLOS on UWB-based positioning. Moreover, the integrated fusion framework effectively improved localization accuracy and robustness, outperforming approaches relying solely on PDR or UWB. These findings underscore the feasibility and efficacy of the proposed method for high-precision pedestrian positioning in complex indoor environments.

Future work may focus on extending the proposed framework to more diverse indoor scenarios, integrating additional sensors, and exploring advanced optimization techniques to further enhance the adaptability and stability of the fusion system.

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